

Joint Detection-Loss Training: MMD vs. No-MMD, Cross-Domain Evaluation

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1 Setup

Unlike every other comparison in this study, these checkpoints were trained with `scripts/joint_train.py`: both domains receive their own detection loss every step (`mmd_cfg.joint_detection_loss=True`), not just the target domain, with the source domain forward-only as in the rest of this study. The MMD variant adds the usual RBF-kernel alignment term (`mmd_weight=0.8`, EMA bandwidth momentum 0.9, `mmd_target_layer=10`, flattened features) on top of that; the no-MMD variant is the joint detection loss alone.

Source = `real_small` or `syn_small` (both project’s smallest subsets, 1600 train images each), target = the other domain, run in both directions. YOLOv10n, `imgsz 1920`, batch 8, 100 max epochs, 2×RTX 4090 via DDP (`--device`) per run. Every resulting checkpoint was then evaluated on *both* val sets – `real_small` val (548 images) and `syn_large` val (640 images, the val split `syn_small`’s own config borrows rather than using its own 160-image val folder) – with person and vehicle reported separately, using a plain `model.val()` call (`imgsz 1920`) independent of which domain that checkpoint was actually trained toward as its target.

2 Results

MMD beats no-MMD on every single row – both classes, both val sets, both directions (8/8). The gap is largest on person in the `syn`→`real` direction evaluated on `real` val (0.530 vs. 0.502 mAP50, +0.028), and smallest – effectively tied – on vehicle/`real-val` in the `real`→`syn` direction (0.854 vs. 0.851). Since every checkpoint here is evaluated on both val sets regardless of which domain was its actual adaptation target, this table also gives a same-model, alternate-domain comparison for free.

3 Compared to source-domain retention under the old (non-joint) regime

Under the standard target-only-detection-loss adaptation used everywhere else in this study, checkpoints score drastically worse on their source domain than on their target domain (see `source_domain_retention.tex`: e.g. mAP50 dropping from the ~0.6–0.9 target-domain range to ~0.02–0.35 on source). Joint training does not show that collapse: every checkpoint above scores in a broadly similar, reasonable range on *both* val sets rather than one domain being near-zero. This is the expected consequence of giving the source domain a real detection-loss gradient every step instead of treating it as a frozen MMD reference – but it has not yet been checked whether this also holds at the project’s standard scale (`real_large`/`syn_large` or larger), only at this small-dataset, 100-epoch, 2-direction dev run.

4 Open items

- Only the project’s smallest subsets (`real_small`/`syn_small`) have been run through joint training so far – no confirmation yet that MMD’s edge over no-MMD, or the retained-both-domains pattern above,

Model vehicle mAP50-95	Direction	Val set	person mAP50	person mAP50-95	vehicle mAP50
0.602 MMD	real→syn	real	0.574	0.263	0.854
	real→syn	syn	0.604	0.294	0.611
	syn→real	real	0.530	0.240	0.831
	syn→real	syn	0.614	0.292	0.612
0.423					
0.582					
0.424					
0.601 no-MMD	real→syn	real	0.571	0.259	0.851
	real→syn	syn	0.588	0.281	0.599
	syn→real	real	0.502	0.227	0.828
	syn→real	syn	0.589	0.264	0.582
0.413					
0.580					
0.401					

Table 1: Every joint-trained checkpoint evaluated on both val sets, person/vehicle separated. Columns are missing an overall/all-class average deliberately – it is a weighted mix of both classes and not the quantity of interest here (as elsewhere in this study).

holds at real_large/syn_large/syn_xlarge scale.

- No comparison yet against this study’s main (non-joint) adaptation regimes at the same source/target pair – i.e. whether joint training’s target-domain numbers are competitive with, better, or worse than the target-only-detection-loss numbers reported elsewhere in this study for the same real_small/syn_small pair.
- Only one seed/run per (variant, direction) cell; no repeated-run variance estimate yet for how much of the MMD-over-no-MMD gap is noise vs. signal.